

Experiment: 7

Analog Beamforming Analysis Using MATLAB

Aim

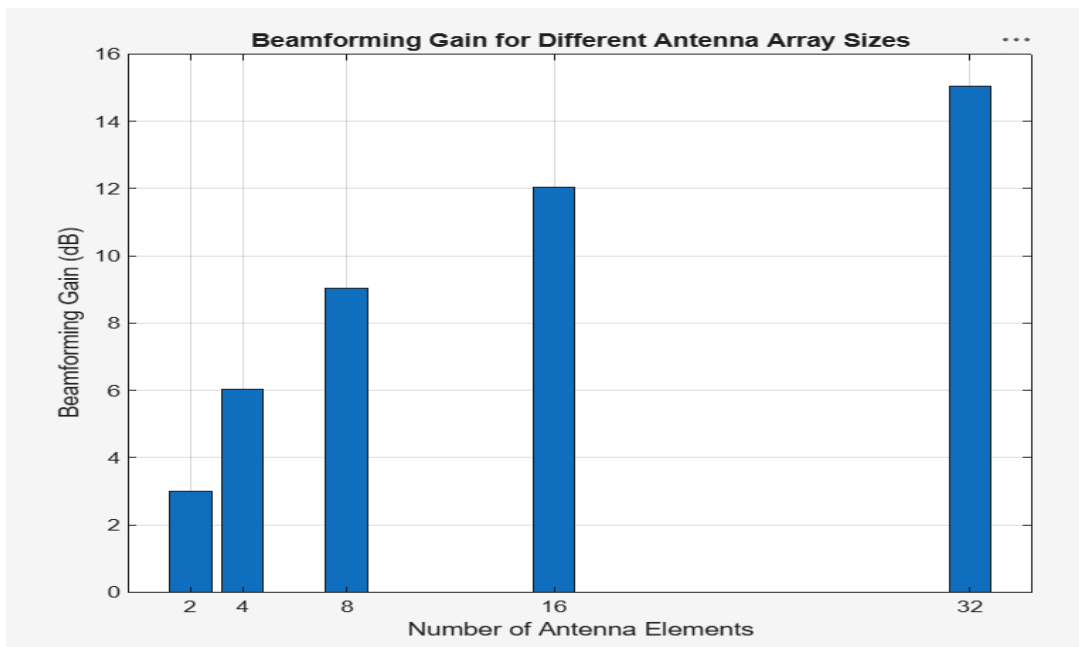
To analyze the performance of analog beamforming by studying beamforming gain, beam width, and main lobe direction using MATLAB.

Objectives

1. To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.
2. To compare the beam width of an antenna array for different beamforming configurations using MATLAB.
3. To analyze the main lobe direction by steering the antenna array beam to different angles using MATLAB.

Objective – 1 Analyze Beamforming Gain (dB)

```
clc;
clear;
close all;
antennas = [2 4 8 16 32];
gain = 10*log10(antennas);
figure
bar(antennas, gain)
grid on
% Number of Antenna Elements
% Beamforming Gain (dB)
xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```

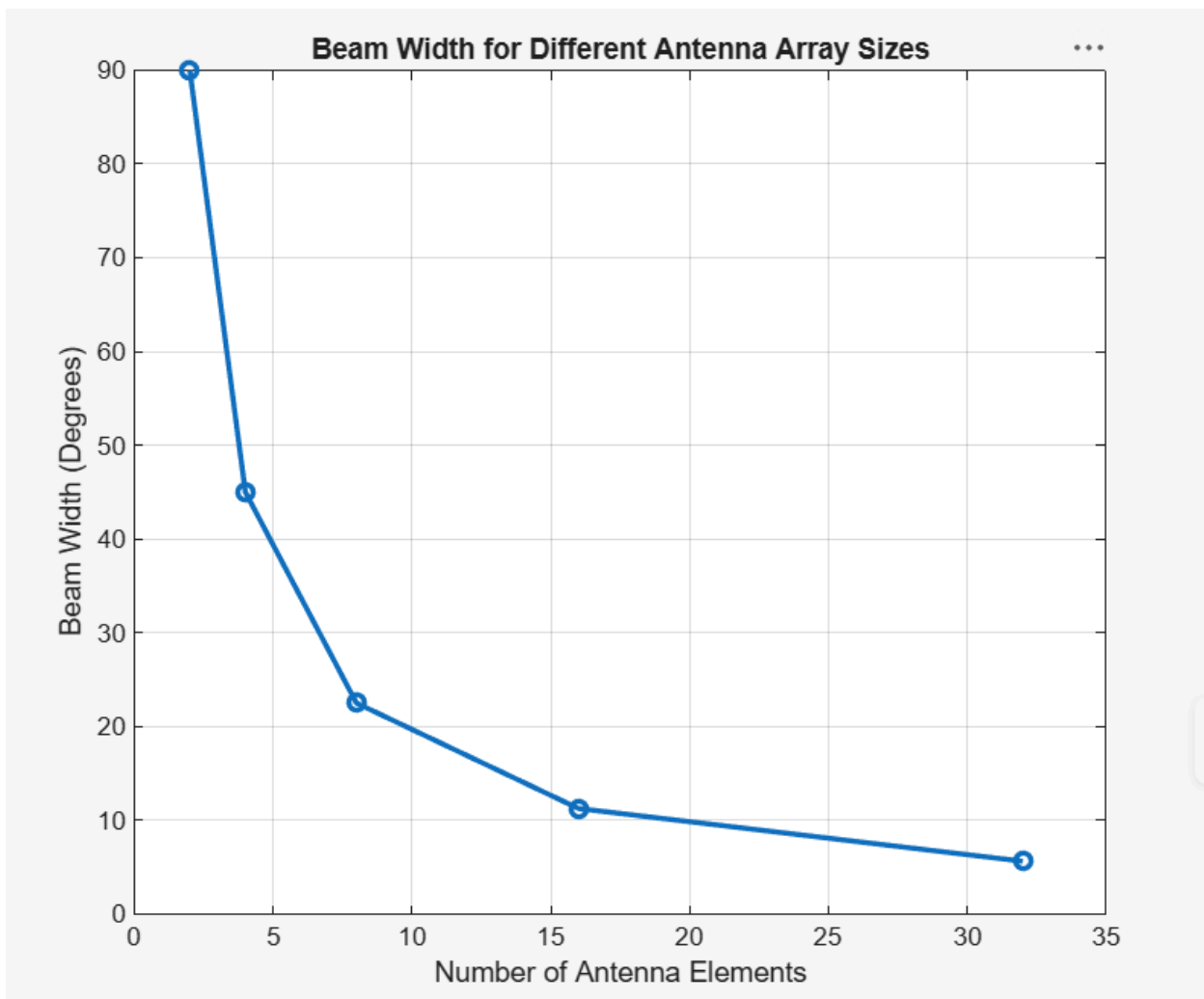


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Objective – 2 Compare Beam Width (Degrees)

```
clc;
clear;
close all;
antennas = [2 4 8 16 32];
% Number of Antenna Elements
beamwidth = [90 45 22.5 11.25 5.625];
figure
% Beam Width (degrees)
plot(antennas, beamwidth, 'o-', 'LineWidth', 2)
grid on
xlabel('Number of Antenna Elements')
ylabel('Beam Width (Degrees)')
title('Beam Width for Different Antenna Array Sizes')
```



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Objective – 3 Analyze Main Lobe Direction (Degrees)

```
clc;  
clear;  
close all;  
theta = [-60 -30 0 30 60];  
gain = [1 1 1 1 1];  
figure  
% Main Lobe Direction (degrees)  
% Constant normalized gain  
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)  
title('Main Lobe Direction in Analog Beamforming')
```

